

Answer key

Form P · Practice

Marking guide and worked solutions · 40 points · may be shared with students after the review

How to mark. Part 1 and the Summary reward **interpretation that is correct, specific and clearly explained in the student's own words**; there is no single right model. A full coefficient interpretation has: the change in X **with its unit** (1 °C, €0.10, 10 km/h); the change in predicted Y **with direction and unit** (more or fewer cups); **"holding the other variables constant"** (naming them); and association language ("is associated with", "the model predicts"), not "causes". For 0/1 variables the comparison group must be clear (event vs no event). Apply **follow-through** from earlier answers; accept rounding differences of ± 0.02 in s and t and ± 0.5 in forecasts. Students interpret coefficients but are **not yet** asked whether coefficients are significant. In Part 3 a t-test is significant if $p < 0.05$ (two-tailed, T.TEST type 2). Objective column: syllabus learning outcome (LO) and your category, **Con** concept, **Cal** calculation, **Com** computation in Sheets.

1 · Quick answer sheet

Q	Objective	Expected answer	Pts	How the marks split
1a	LO1, LO5 · Con	Any three sensible inputs with a direction and a reason: weekend (up), beach event (up), planned price (down), temperature (either direction with a reason; the data show down), wind (down: fewer outdoor customers).	3	1 per input: direction and a coherent reason; either direction for temperature scores if defended.
1b	LO5 · Con	Item 7 , gelato sales counted at closing time: it is not known before opening and it is driven by the same weather and day type.	2	1 item 7 + 1 the problem.
1c	LO3, LO5, LO6 · Con	Not both. $r = 1.00$: they are the same information (the price changed once, on the day the flag switches), so the model cannot separate them. Keep the price (the thing the café controls) or the flag, not both.	2	1 identical information and why it is a problem + 1 a choice.
2a	LO5, LO6 · Com	Sales = $84.76 - 0.81 \times \text{temperature} - 0.18 \times \text{wind} + 11.53 \times \text{weekend}$ ($\hat{y} = b + m_1x_1 + m_2x_2 + m_3x_3$; Table 2: Col A = m_3 , Col B = m_2 , Col C = m_1 , Col D = b).	2	1 numbers matched to the right variables + 1 equation form with variable names.
2b	LO6, LO7 · Con	(i) Holding wind and weekend constant, a 1 °C rise in temperature is associated with about 0.81 fewer pastries sold per day. (ii) The other variables also change sales and may differ between hot and cooler days. Holding them constant isolates the effect of this variable: we compare days that are alike in the other variables and differ only in this one.	3	(i) 1 size, direction and unit + 1 names the variables held constant; (ii) 1 reason.
2c	LO6, LO7 · Con	(i) Negative: warmer days go with fewer pastries (about 0.8 per °C, holding wind and weekend constant). A possible reason: customers switch to cold items (gelato, iced coffee) in the heat. (ii) Weekends sell about 11.5 more pastries than weekdays, at the same temperature and wind. (iii) $10 \times (-0.18) = -1.8$: about 1.8 fewer pastries per 10 km/h of extra wind.	3	1 sign meaning with a reason + 1 0/1 variable with the comparison group + 1 rescaled coefficient.
2d	LO3, LO6, LO7 · Cal + Con	(i) $10 \times (-0.81) = -8.1$: about 8 fewer pastries per 10 °C. (ii) "Per 10 °C" is better: 1 °C is too small to picture and hot and cool days differ by 10 °C or more. (iii) Not fair: weekend is a 0/1 switch (+11.5) and wind is per km/h; for a realistic 10 km/h change wind is -1.8, so weekends matter about 6 times as much (not 64).	3	1 rescaled + 1 why a sensible unit + 1 unfair comparison explained with 2c(iii).
2e	LO5, LO6, LO7 · Cal + Con	(i) $\hat{y} = 84.76 - 0.81(20) - 0.18(12) + 11.53(1) = 77.9$ pastries. (ii) $10 \times (-0.18) = -1.8$ pastries.	3	2 prediction + 1 wind change effect with sign.
3	LO3, LO4, LO6 · Con	(a) Day 4 : sales far above the others at a middling temperature. (b) The slope barely changes (-1.14 with it, -0.98 without) but R^2 jumps from 0.26 to 0.96: the spike spoils the fit, not the trend. (c) Find out why (a festival tie-in, a promotion, a recording error). Delete only with a documented reason; otherwise keep it, flag it and report both.	3	1 + 1 (slope and R^2 both) + 1.
4a	LO3 · Cal	Means 66 and 63; $\Sigma(x - \bar{x})^2 = 40$ and 96; $s_1 = \sqrt{(40 \div 4)} = 3.16$, $s_2 = \sqrt{(96 \div 4)} = 4.90$ (deviations in the worked table below).	3	1 means + 1 sums of squares + 1 s values.
4b	LO3, LO6 · Con	(i) Group 2 (after) is more spread out ($s = 4.90$ vs 3.16). (ii) Larger standard deviations mean the daily sales bounce around more, so a gap between the two means is easier to explain by chance and harder to detect.	2	1 + 1.

5a	LO3, LO5 · Com	(i) =STDEV.S(D2:D6) (or STDEV) · (ii) =T.TEST(D2:D6, E2:E6, 2, 2) · (iii) =LINEST(G2:G169, H2:J169, TRUE, TRUE). Sheet names are not required.	3	1 each (correct function and ranges; arguments in the right order).
5b	LO3, LO5 · Com	In T.TEST the first 2 means a two-tailed test and the second 2 means two samples with equal variance . In LINEST the first TRUE keeps the intercept (not forced to 0); the second TRUE returns the extra statistics (standard errors, R ² , SE – Y, F, df, sums of squares) in rows 2–5.	2	1 T.TEST arguments + 1 second TRUE.
5c	LO1, LO3, LO6 · Con	(i) H ₀ : the price rise made no difference to mean daily pastry sales. (ii) p = 0.283 > 0.05: not significant : no clear evidence of a difference. (iii) No. “Not significant” means the data do not show a clear difference; it does not show there is none (absence of evidence is not evidence of absence). With only 5 + 5 days a real drop of a few pastries could be missed.	3	1 null hypothesis + 1 decision using p + 1 absence of evidence ≠ evidence of absence.
6	LO7 · Con	(i) One model input with a sensible change and its effect on sales, holding the others constant (e.g. “Holding wind and weekend the same, a 10 °C warmer day goes with about 8 fewer pastries.”). (ii) The price test in plain words: pastry sales were a little lower after the price rise but the difference is not significant (about 3 pastries, p = 0.283). (iii) A caution (small samples, other changes, association not proof, forecasts can miss).	3	1 + 1 + 1; plain language and correct units.

2 · Worked table (Part 2)

Day	Before (x ₁)	x ₁ – $\bar{x}_1$	(x ₁ – $\bar{x}_1$) ²	After (x ₂)	x ₂ – $\bar{x}_2$	(x ₂ – $\bar{x}_2$) ²
1	66	0	0	61	–2	4
2	70	+4	16	69	+6	36
3	62	–4	16	57	–6	36
4	68	+2	4	67	+4	16
5	64	–2	4	61	–2	4
Sum	330	0	40	315	0	96
Mean $\bar{x}$	66			63		
s	3.16			4.90		

Check against Sheets: =T.TEST gives p = 0.2832 (df = 8, |t| = 1.150, two-tailed).

3 · Where students go wrong

- **Leaving out “holding the other variables constant”** (2b, 2c, Q6), or not naming what is held constant.
- **Treating a coefficient as the total effect of a big change** (2c(iii), 2d): the coefficient is per **one unit**; students must multiply by the change they mean (€0.10, 10 °C, 10 km/h).
- **Sign confusion**: a negative coefficient means sales **fall** as X rises; it does not mean negative sales or “no sales”.
- **Comparing raw coefficients across variables with different units** (2d(iii)).
- **Reading the LINEST table backwards** (2a): Col A is the **last X** column and Col D the intercept. The layout table is printed on the paper, so give no credit for guessing the order; apply follow-through if the order is wrong but the method is right.
- **“Causes” and “per customer”**: the coefficients describe daily totals and associations.
- **Reading p as the probability that H₀ is true** (5c). Credit “if there were no real effect, a result this extreme would be rare”.